Chunqiu Steven Xia — Teaching Statement

I am excited and interested by the opportunity to serve as a teacher and mentor for the next generation of students. During my PhD, I was a teaching assistant for a graduate-level course focused on software engineering and Generative AI (GenAI) in which I designed brand new course assignments for and gave lectures on key topics. I gave multiple guest lectures at UIUC, other universities (such as North Carolina State and Ohio State University), and industry research groups. Additionally, I had the privilege of mentoring eight students (two junior PhD, three masters, and three undergraduate) on their research. Teaching is both illuminating and deeply fulfilling to me. I take great joy in helping students strengthen their skills and pursue their aspirations. Furthermore, I value the insights I gain from them in return.

Teaching Philosophy

My teaching philosophy is guided by my past experiences as both a student and teacher and consists of the following principles: (1) **active learning through engaging and interactive classroom experiences**; (2) **practical and hands-on assignments and projects to develop problem solving skills**; (3) **academic environments that encourages self-learning**. Research has demonstrated the positive effect of active learning activities in improving student interest and learning abilities [1]. I aim to incorporate active learning concepts into my lectures with interactive, discussion-based activities. I also believe in the value of hands-on learning especially in computer science. As such, I plan to supplement classroom teaching with challenging assignments to help students develop not only problem solving skills but also experiences with the latest technology. I strive to build a classroom setting that facilitates learning by encouraging questions, discussions, and feedback.

Teaching Experience

My teaching experiences as a graduate student come from: (1) I was a teaching assistant for a graduate-level software engineering course. The course is on the intersection of GenAI and software engineering where I designed the homework assignments and gave lectures on key course topics. (2) I gave guest lectures on software engineering concepts for multiple courses, including ones in other universities (such as North Carolina State and Ohio State University).

Course homework design. The course I TAed was a brand new course in terms of course materials with the topic being on the intersection of GenAI and software engineering. As such, I created completely new homework assignments with the aim to bridge the gap between the concepts taught in lecture and practical technology that can be used by the students in their own research or careers. I created in total four assignments, ranging from program repair to vulnerability detection, where each assignment involves building a solution to a challenging software engineering problem using tools and concepts from GenAI. Given the nature of GenAI where credits or GPUs are required for the usage of state-of-the-art models, I designed assignments on Google Colab[1] to facilitate the easy usage of necessary resources and reduce the barrier for students to complete their homework assignments. Furthermore, I also included related or additional paper readings for each assignment to encourage and facilitate further learning for students interested in the homework topic. Due to the success and originality of the course, professors from other universities (such as Cornell University) have asked to use the created homework assignments as references for their own courses.

Course lectures and student interactions. In addition to creating homework assignments, I was also actively involved in teaching the students on course materials. In particular, I presented a lecture on the basics of GenAI and large language models (LLMs). As the course is graduate-level course attended by students across multiple domains who may not all be well-versed in GenAI, I prepared the lecture from the ground up by starting from simple concepts and explaining each key concept with examples. Furthermore, I connected the overall lecture to how GenAI may be used for software engineering tasks by showcasing not only recent research breakthroughs but also exciting industry developments in this area. Overall, the feedback and response from the students was positive as they had several interesting questions and remained engaged throughout the lecture. Another part of my duty was responding to student questions and concerns on the communication platform Campuswire. My goal is to provide timely and informative replies to student questions and as the semester continued, I started to see more and more

[1] https://colab.research.google.com/

students engaging on the platform to obtain answers for their questions and also to interact and help other students. Additionally, I allocated extra time especially prior to assignment or project deadlines as I noticed there was an increase in questions in order to properly address any time-sensitive concerns. I also met with individual students to provide additional help whenever online discussion was insufficient. My efforts have also been recognized by students in course evaluations, where I received overwhelmingly positive reviews, placing me within the Top 1% of TAs in the entire department.

Guest lectures. I gave two guest lectures on software engineering topics related to my research area for courses at other universities (such as North Carolina State and Ohio State University). I also gave multiple presentations and lectures focused on specific research projects for industry research groups. During my lectures, I start by illustrating the background and theory of the topic to students through practical and easy to digest examples. For example, when I gave a lecture for program repair, I used the same code snippet to first show how a potential bug might appear, what a naive patch could look like, and finally how advanced techniques can generate correct patches. Through these examples, the students can understand the key materials of the lesson and remain engaged. I also encouraged student interactions by designing challenging scenarios and asking them to come up with solutions. I found that these interactions can reinforce previously taught concepts and identify which parts of the lesson students struggle with.

Advising

During my PhD, I had the pleasure of mentoring and advising several junior students. In total, I worked closely with and mentored two junior PhD, three masters, and three undergraduate students across a wide range of different backgrounds and experience levels. My mentees and I submitted seven papers to top-tier software engineering and machine learning conferences, all of which were successfully published. Furthermore, three of my mentees have gone on to begin their own PhD studies.

I found my advising and mentoring experiences incredibly fulfilling. My mentoring and advising approach typically starts by allowing them to explore their own individual research interests within the topics that I am working on. For students who are beginning their research careers, I adopt a more hands-on approach by designing specific, concrete tasks that they can work on. This allows students to learn the basics of a research project, contribute meaningfully to the project, and gain confidence to develop their own research ideas. For more experienced students, I prefer to let them explore their topics of interest by providing feedback on their projects, introducing relevant prior work to them, and assisting them in designing or implementing various project details. I encourage constant discussion with my mentees regarding any difficulties or challenges they may encounter. This is also something I learned from my own advisors and has been incredibly helpful for me. Rather than optimizing for the completion of a particular project, I aim to help students to become better researchers and develop their own research vision.

Teaching Interest

I am interested in teaching topics relevant to software engineering, including classes on software design, testing and analysis, programming languages, and security at the undergraduate level. I aim to prepare students with a solid theoretical foundation as well as the ability to transfer these foundations between different programming languages, frameworks, and domains.

At the graduate level, I hope to develop a course on the intersection of software engineering and GenAI. In this course, I want to introduce students to recent developments in both research and industry in utilizing GenAI tools to solve software engineering problems. I hope to connect traditional software engineering topics like test generation or code synthesis with GenAI developments like LLMs and agentic systems, allowing students to learn cutting-edge techniques and insights that can inspire their own research. Additionally, I would like to organize seminars for discussing recent research and advancements in software engineering and other closely relevant fields like natural language processing, programming languages, computer security, systems, computer architecture, and HCI.

References

[1] Scott Freeman, Sarah L Eddy, Miles McDonough, Michelle K Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the national academy of sciences*, 2014.